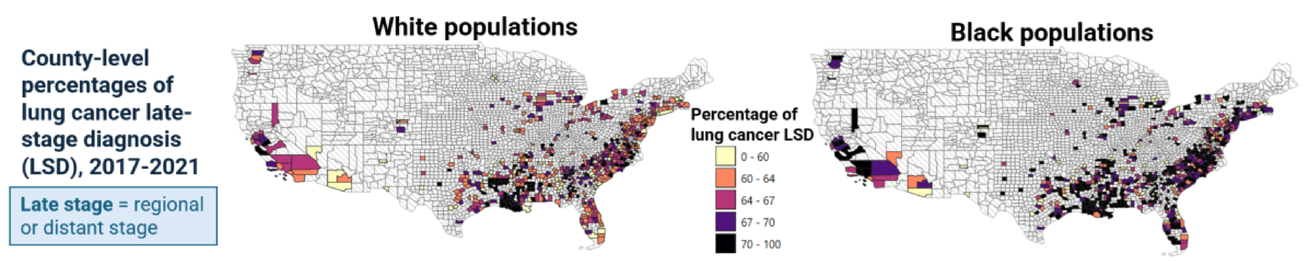
Shifting the Effectiveness-Equity Frontier Outward: A Causal AI Framework to Prioritize Synergistic Interventions in Improving County-Level Lung Cancer Late-Stage Diagnosis Outcomes

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Introduction

- Only 28.1% of U.S. lung cancers are diagnosed at an early stage.
- Black individuals experience a higher burden of late-stage diagnosis (LSD) and are approximately 13% to 16% less likely to be diagnosed early than White individuals.
- These disparities reflect persistent structural and systemic inequities in access to screening, healthcare resources, and timely diagnostic care.
- Identifying the key drivers of LSD and racial disparities is critical for designing targeted, equitable interventions.



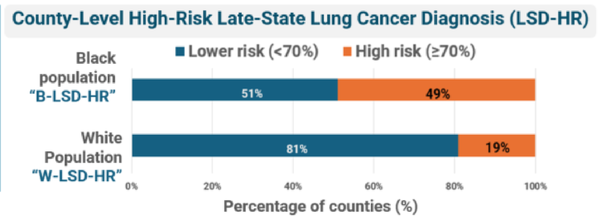
Goal: Develop a data-driven causal discovery and reasoning framework to identify actionable interventions for reducing high lung cancer LSD rates and Black-White disparities at the county level.

Methods

Analysis Data

662 U.S. counties | 80 county-level factors | 6 Domains | 2 Outcomes (B-LSD-HR & W-LSD-HR)

Factor Domains: demographics, social determinants of health, community health indicators, structural and environmental factors, lung cancer screening, and healthcare access.



Key Equations for Intervention Analysis

1 Backdoor-adjusted intervention probabilities

$$P_{g,j}^{INT} = P(Y_g = 1 \mid do(X_j = x_j)) = \sum_{z} P(Y_g = 1 \mid X_j = x_j) P(Z_j = z)$$

INT: intervention; j: intervention scenario; X_j: intervention factor(s); x_j : assigned level(s); Y_g: group-specific binary outcome; Z_j: adjustment set; g: Black (B) or White (W) group.

Backdoor adjustment used to estimate intervention-specific probabilities for B-LSD-HR and W-LSD-HR.

2 Effectiveness and Equity measures

B-LSD-HR improvement	$\Delta B_j = P_B^{SQ} - P_{B,j}^{INT}$
W-LSD-HR improvement	$\Delta W_j = P_W^{SQ} - P_{W,j}^{INT}$
Mean improvement	$M_j = (\Delta B_j + \Delta W_j)/2$
Black-White (B-W) gap reduction	$R_j = \Delta B_j - \Delta W_j$

SQ: status quo; INT: intervention; j: intervention scenario.

Status quo: $P_B^{SQ} = 48.39\%$,; $P_W^{SQ} = 19.66\%$; B-W gap= 28.73 %

3 Preference-weighted composite score

$$Score_j = W_E M_j + W_Q R_j$$

$$W_E + W_Q = 1$$

W_E : effectiveness weight; W_Q: equity weight

Preference settings:

$(W_E, W_Q) = (1.00, 0.00), (0.75, 0.25), (0.50, 0.50), (0.25, 0.75), (0.00, 1.00)$

4 Synergy of combined interventions

Two-factor combination:

$$S_{AB} = Score_{AB} - Score_A - Score_B$$

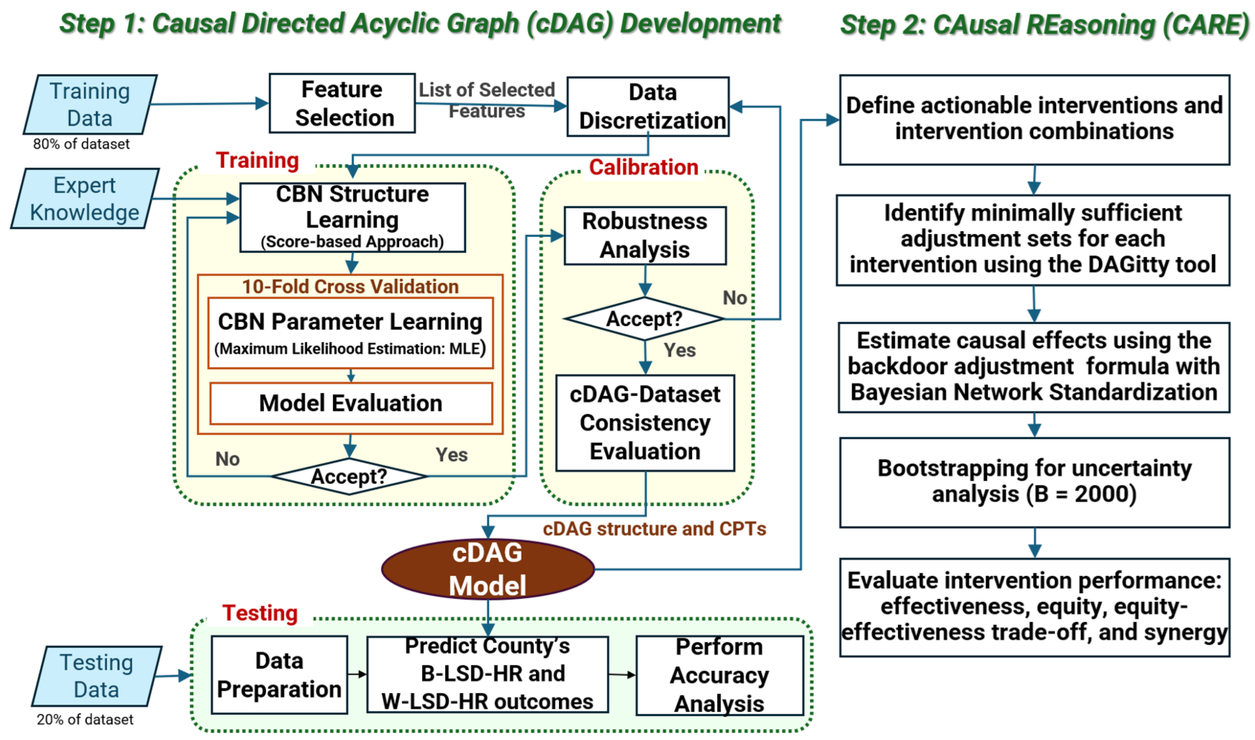
Three-factor combination:

$$S_{ABC} = Score_{ABC} - Score_A - Score_B - Score_C$$

S: departure from single-factor additivity; $Score_A, Score_B, Score_C$: preference-weighted scores for single-factor interventions; $Score_{AB}, Score_{ABC}$: scores for combined interventions.

$S > 0$: greater benefit than expected under additivity; $S = 0$: consistent with additivity; $S < 0$: less benefit than expected under additivity.

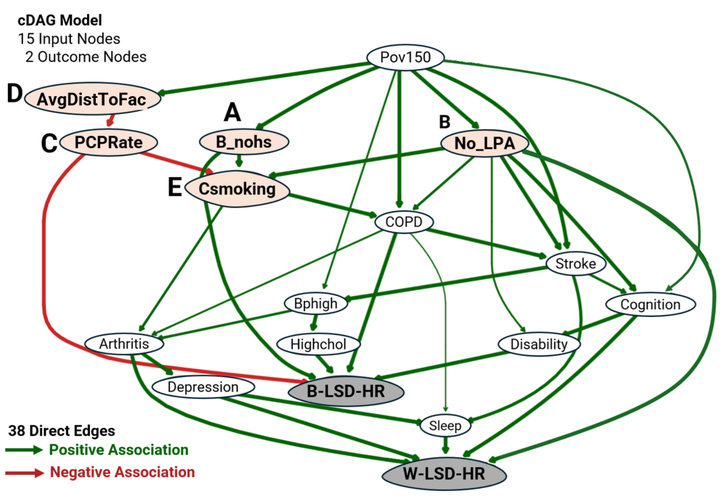
Causal Directed Acyclic Graph and Causal Reasoning (cDAG-CARE)



Abbreviation: CBN = Causal Bayesian Network; CPTs = Conditional Probability Tables

Results

Part 1: The Calibrated cDAG Model



Node	Definition/ Intervention Level
B_nohs (A)	Percentage of Black adults aged ≥25 years without a high school diploma
	B_nohs = Low (≤ 12%)
No_LPA (B)	Percentage of no leisure-lime physical activity among adults aged ≥18 years
	No_LPA = Low (≤ 24.5%)
PCPRate (C)	Ratio of population to primary care per 100,000 population
	PCPRate = High (≥ 288 PCPs/ 100,000)
AvgDistToFac (D)	Average distance to closest screening facility
	AvgDistToFac = Close (≤ 7.25 Mile)
Csmoking (E)	Percentage of current smoking among adults aged ≥18 years
	Csmoking = Low (≤ 14.1%)

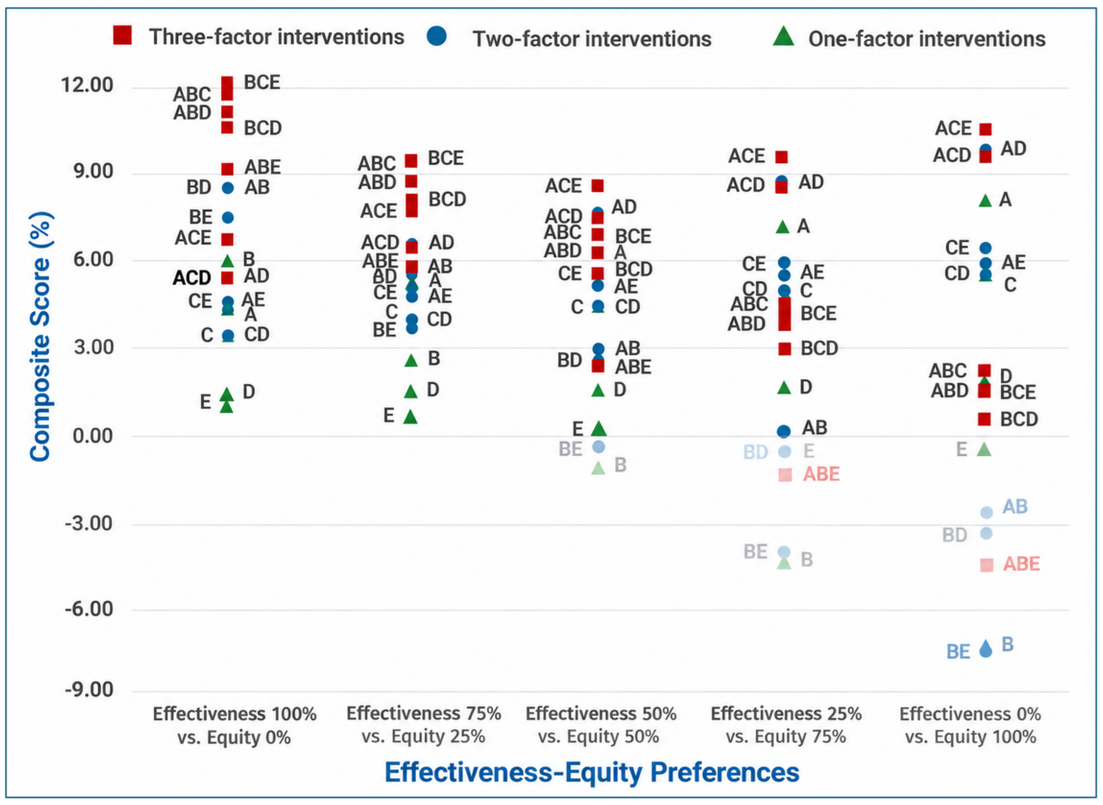
Model performance: The area under the free-response receiver operating characteristic curve (AU-FROC) was 0.79 (95% CI: 0.76, 0.82) in training and 0.75 for testing.

Candidate interventions: Five actionable factors (A-E) yielded 25 scenarios, (5 single-, 10 two-, and 10 three-factor).

Causal estimability: Of 25 scenarios, 22 had valid adjustment set for causal effect estimation.

Non-estimable scenarios: The causal effects of three scenarios (DE, ADE, and BDE) with no valid adjustment set are not identifiable.

Part 2: Effectiveness–Equity Evaluation of Actionable Interventions based on Causal Reasoning on the cDAG

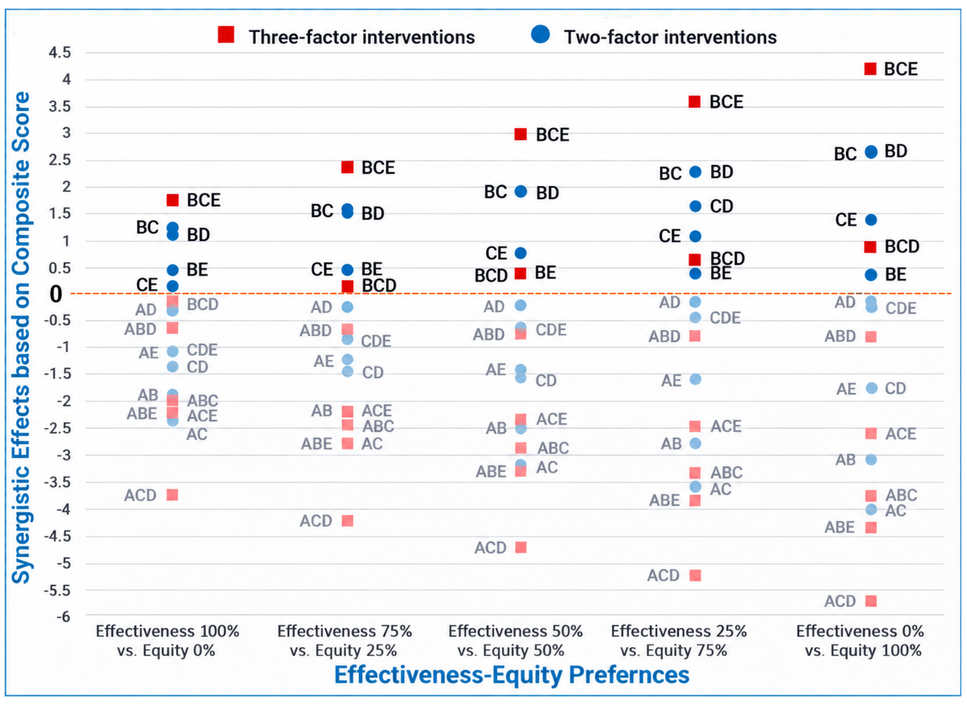


Key findings:

- All 22 estimable interventions improved both B-LSD-HR and W-LSD-HR, but not all reduced Black-White disparities.
- **Single-factor:** "B" ranked highest for effectiveness; "A" for equity and balanced preferences.
- **Two-factor**: "AB" ranked highest for effectiveness; "AD" for equity and balanced preferences.
- **Three-factor:** "BCE" ranked highest for effectiveness; "ACE" for equity and balanced preferences.

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Part 3: Synergistic Effects of Combined Interventions Across Effectiveness-Equity Preferences



Key findings:

- "BCE" showed the strongest synergy, consistent with interconnected barriers represented in the calibrated cDAG model.
- Synergy may allow lower-intensity intervention components, though cost-effectiveness will be evaluated.
- Synergy can evaluate interactions to identify redundancy and avoid ineffective combinations, such as "ACE".
- Coordinated interventions targeting complementary causal pathways may shift the effectiveness–equity frontier outward.

Conclusions

- We developed a cDAG-CARE framework to identify and monitor Black and White groups' lung cancer LSD-HRs and their disparity with potential actionable interventions.
- The causal AI framework can provide effective, efficient, and equitable synergistic interventions based on leisure time, primary care, and current smoking status to improve LSD outcomes at the US county-level.

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